

1. What are the biggest barriers to private sector innovation in AI for health care and its adoption and use in clinical care?

Practically, buy-in from providers to use new AI tools, their confidence in their efficacy, and their skepticism that the tools are just going to be “another expense.” As a patient and parent, I have seen some AI tools been positively received by providers. For example, my sons’ endocrinologists use AI scribes regularly to support notetaking and follow up (both my boys have Type 1 diabetes). So, these types of tools have helped our medical professional community become comfortable with AI, but future innovations have to take small steps forward, I believe, for most providers to embrace them.

Providers believe in rigorous evidence; they are trained on the randomized clinical trial being the gold standard for care. And, then in practice, they collect data through every observation they make with their own patients. For AI to flourish in support of clinical care, providers will need to feel confident that the “algorithms” behind the AI are based on appropriate data. At first, they will want to vet that the “answers” or “support” they are receiving is accurate and in line with their own experience before beginning to trust it. Gaining providers’ trust will be one avenue to getting their buy-in that the cost of any new tools are worth it and not just another expense.

Policymakers may be able to facilitate the use of AI tools by supporting independent assessments of tools in a sort of clearinghouse. Or, maybe less costly, by identifying the preferred characteristics and operational parameters required for high quality tools used in clinical care. Some of this information can come from providers themselves who would be on the front lines working with tools.

4. For non-medical devices, what are the most promising AI evaluation methods (pre- and post-deployment), metrics, robustness testing, and other workflow and human-centered evaluation methods for clinical care? Should HHS further support these processes? If so, which mechanisms would be most impactful (e.g., contracts, grants, cooperative agreements, and/or prize competitions)?

Assessments of AI for clinical care could be rooted in an implementation science framework. The question asked specifically, and appropriately, about workflow which is an underlying and assumed benefit of the injection of AI into clinical care. That is, the use of AI by clinicians should, hopefully, ease their workflow and allow for more meaningful interaction with patients. The promise of the availability of data from the use of AI agents in clinical care offers a unique opportunity to assess the implementation of this technology. With much less burden than in other research before it, we could answer questions like: How do providers use the tools available? Is the information from the tools accurate? Do providers continue to use the same tool for similar patients or does their use drop off over time? Is the practice of care for specific conditions different with the tools and in what ways?

The deployment of new tools into clinical practice could allow for very robust testing in short amounts of time. It would be important, however, that testing occur in a multitude of different and varied locations to understand how different providers in different settings might use the new technologies. The CMS Innovation Center, for example, could consider a model that supports the use of new technologies in a tiered approach. First, by soliciting organizations that sell such AI tools to participate and put their tools “to the test” and second to solicit providers and provider groups to participate. It may have to offer further incentives to providers in some settings (e.g., rural communities). CMS could also request large health systems that already use in-house AI tools to also participate with their own technologies. There is some precedence for this type of approach of “putting your approach to the test” with the test of disease management programs by CMS from 2003 to 2008. There are ways to evaluate these tools without the need for randomized controlled trials or “usual care” comparison groups as well. Any assessment of tolls might also consider patient-facing tools that support providers and patients as well.

5. How can HHS best support private sector activities (e.g., accreditation, certification, industry-driven testing, and credentialing) to promote innovative and effective AI use in clinical care?

The development of standards for industry-testing could go a long way to support providers’ buy-in and confidence that AI tools are efficacious. It may be resource-intensive, however, to go as far as accreditation and certification and only companies that have available resources (or are able to charge users a premium) may be to afford these steps.

8. Where would enhanced interoperability widen market opportunities, fuel research, and accelerate the development of AI for clinical care? Please consider specific data types, data standards, and benchmarking tools.

For this question and the next two, I will lean heavily on my experience as a father of children with Type 1 Diabetes and a son with a father and mother-in-law with the same disease. For many individuals living with T1D, there is a lot of data available that is not always easy accessible to them or their providers and certainly not integrated with other information. For example, many individuals with T1D (though nearly not enough) use continuous glucose monitors (CGMs) to track their blood sugar continuously but not many providers or patients themselves have access to the data (which is collected in 5-minute intervals). Freeing these data up for use by patients who may, say, use AI algorithms to support their care and optimize their insulin dosages could be a major step forward for people living with T1D. The highs and lows that come with less than accurate insulin dosing are a frequent cause of stress and concern for these patients and their caregivers and, too often, result in concomitant mental health conditions as

well. Those patients who additionally use an automated insulin delivery device (casually referred to as an insulin pump) could also then bring together information on insulin dosage with their blood sugar levels to support stronger control. There are already non-FDA approved, community-built apps for this purpose (for example, Trio) that are moving in this direction but the case of T1D could be a promising pilot opportunity for the federal government given the prevalence of T1D in the United States and its annual incidence rate. (My children or other family members do not use this app but we have met families that do.)

9. What challenges within health care do patients and caregivers wish to see addressed by the adoption and use of AI in clinical care? Equally, what concerns do patients and caregivers have related to the adoption and use of AI in clinical care?

As a caregiver, a significant concern that I have is reliability of the potential AI technology. Taking the case of T1D, for example, it is a disease that can be very challenging and frustrating to manage. Until we have cell therapy cures, there is no “silver bullet” solution. Speaking as a father of two teenagers with T1D, each one of my boys is completely different in his reaction to insulin and the same carbohydrate-laden foods. This is typical of the community of individuals with T1D. So, as a parent, I face this dichotomy. On the one hand, I desperately would love for someone to harness the power of AI/agentive AI/etc. to develop a tool to support my children so that some stress is relieved from them and my wife daily on making decisions. It is established that people with T1D make 180 to 300 additional health-related decisions a day.¹ We would love a tool to relieve the stress of making these decisions. On the other hand, I would be cautious about any tool that purported to be a solution without knowing that it was tested and vetted in a standardized and rigorous fashion. So, my concern? Can we have an efficient process to vet tools that is timely, is patient-centered, and cost-efficient so that we can provide families confidence that the tools being rolled out into any clinical care setting are safe and reliable?

10. Are there specific areas of AI research that HHS should prioritize to accelerate the adoption of AI as part of clinical care?

Please, consider prioritizing the Type 1 diabetes space for exploration of AI research. While my comments above have focused on individuals with T1D, I would also implore that just as important an area are those individuals who have evidence of the antibodies for T1D but are not yet insulin dependent. We have the technology and ability today to screen for all the antibodies that science has proven result in a degradation of one's beta cells. One of my sons was diagnosed as such and we monitored him for multiple

¹ [The stress of making diabetes decisions](#)

years before he became insulin dependent. Unfortunately, not everyone who could develop T1D is tested. And, more unfortunately, “crashing into” T1D is a very traumatic event for people with this disease and their loved ones.

In addition to supporting individuals who are currently insulin dependent, AI research could also support our knowledge of whom might be most likely to have antibodies, could support individuals who have these antibodies, and support providers and caregivers. The amount of information for individuals who are about to become or are insulin-dependent has proliferated and making sense of it for each individual case can be overwhelming. For both patients and providers (many of whom do not know a lot about T1D), education modules tied to AI could support better learning and the stress of this disease.